

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Computer Science and Engineering	IIT Gandhinagar	2023-2027	8.30/10
Class XII	Physics, Chemistry, Maths	White Leaf Public School	2022-2023	94.40%
Class X	—	Modern Delhi Public School	2020-2021	96.60%

INTERNSHIPS

- Software Engineering Intern, Intentyfi

[February '26 - July '26]

◦ Architected and built the **Case Management Application (CMA)** from scratch using **Rust, Axum, PostgreSQL, Next.js, TypeScript, and Tailwind CSS**, enabling case monitoring, reviewer workflows, dashboards, and Human-in-the-Loop (HITL) oversight.

◦ Built production **RAG infrastructure** covering document extraction, configurable chunking, Gemini embeddings, PostgreSQL vector storage, and similarity retrieval for enterprise knowledge grounding.

◦ Implemented an end-to-end **HITL pause/resume protocol** across agent runtime and case-management services, supporting Approve, Reject, Escalate, and Reopen reviewer workflows with auditable state transitions.

◦ Implemented AI-agent security controls including **PII masking, multi-layer prompt-injection detection, threat categorization, audit logging, and BYOK credential management**.

◦ Built **AST-based SQL validation** using Rust's `sqlparser`, enforcing single-statement read-only queries and preventing DDL/DML execution from LLM-generated or user-submitted SQL.

◦ Engineered LLM orchestration systems using **Python, FastAPI, Google ADK, and Gemini**, including schema grounding, conversation compression, context-aware follow-up suggestions, asynchronous BI execution, result caching, and artifact scoping.

◦ Implemented production frontend applications using **Angular, RxJS, Next.js, TypeScript, and Tailwind CSS**, including dashboards, database explorers, API/MCP tooling, governance interfaces, and case-management workflows.

◦ Implemented cloud deployment and infrastructure workflows using **GCP Cloud Run, Cloud Build, Cloud SQL, Vertex AI, Docker, and GCP IAM**; integrated token metering and billing workflows using Rust/Axum and PostgreSQL.
- Founder's Office - Technology, Aback.ai

[June '25 - February '26]

◦ Partnered directly with founders to lead technical architecture, product engineering, and automated workflows across multiple early-stage AI products from conception to production deployment.

◦ Delivered software systems spanning **finance, inventory, recruitment, and sustainability**, translating business requirements into scalable multi-tenant architectures and automated pipelines.

◦ Built and launched **AbackTools.com**, an AI-powered tools platform providing utilities across data processing, image manipulation, PDF workflows, and network diagnostics, generating **~500 clicks/day**.

◦ Contributed to **QRliee**, developing QR-based digital workflows, real-time event analytics, and AI-assisted data-import modules for digital menus, business cards, and building directories.

◦ Built core modules for **Invoice Management Software, Inventory Systems, Aback Terminal, and AI Recruitment Portal**, streamlining enterprise operations and automated candidate screening.
- Development Intern, Trado (Windigo Trade Pvt. Ltd.)

[November '25 - January '26]

◦ Developed Trado's algorithmic trading platform using **LEAN**, the open-source quantitative trading engine by QuantConnect, adapting its architecture to company-specific infrastructure and deployment requirements.

◦ Worked with LEAN's **factory-based architecture** to customize core components for backtesting and live algorithm execution.

◦ Built an end-to-end pipeline enabling transition between **historical backtesting and live algorithmic trading**.

◦ Implemented and integrated brokerage plugins supporting additional **Indian discount brokers**, extending platform compatibility and execution flexibility.

◦ Solely led development of **Connect by Trado** from scratch, building brokerage integration and infrastructure for automated algorithmic trading.
- Development Intern, Curlsek AI Technologies Pvt. Ltd.

[March '25 - July '25]

◦ Developed the core Proof-of-Concept (PoC) for Curlsek.ai's AI-powered cybersecurity portal by building AI agents to detect vulnerabilities such as **SQL Injection and Path Traversal** and supporting automated security testing workflows.

◦ Designed and implemented secure authentication and authorization APIs using **Spring Boot and MongoDB** for the platform's backend.

◦ Built scalable services supporting **user authentication, role-based access control (RBAC), and secure session management** across endpoints.

- Contributed to the early-stage architecture of AI-agent-based offensive security workflows for automated vulnerability analysis and continuous penetration testing.

RESEARCH PROJECTS

- **AI in Designing Effective Biomaterials** [August '26 - Present]
Prof. Mukesh Dhanka, IIT Gandhinagar
 - Architected a full-stack platform predicting biopolymer-solvent interactions and material properties (T_g , dielectric constant, Flory-Huggins χ , solubility) under varying conditions (pH, temperature, concentration).
 - Engineered molecular featurization pipelines using **RDKit**, extracting 1024-bit Morgan fingerprints and 2D physicochemical descriptors from polymeric PSMILES representations with automated validation and deduplication.
 - Trained multi-seed **XGBoost ensembles** with uncertainty quantification (σ), k-NN applicability domain (OOD) estimation, and Bemis-Murcko scaffold splits to prevent data leakage and ensure generalization.
 - Implemented high-performance multi-service architecture: a **Rust (Axum)** orchestration API with **PostgreSQL/pgvector** literature RAG retrieval, a **FastAPI** inference engine, and a **Next.js** analytics interface.
- **Chunking in Chess using Python** [January '25 - April '25]
Prof. Krishna Prasad Miyapuram, IIT Gandhinagar
 - Defined cognitive move chunks using **Elapsed Move Time (EMT)** signals extracted from PGN archives, operationalizing the initial think-time pause as the boundary between sequential chunks for players rated 2000+ on Lichess.
 - Implemented a novel game-phase segmentation algorithm comparing **Moving Average, EMA, Standard Deviation, Coefficient of Variation, K-Means clustering, KNN, Random Forest, SVM, and ARIMA** methods to demarcate opening, middlegame, and endgame boundaries from pooled EMT data.
 - Correlated chunk-size distributions and EMT patterns against player **Elo ratings**, cross-validated with **Stockfish centipawn-loss and win-probability** evaluation to measure decision quality across game phases.

TECHNICAL PROJECTS

- **YourCode - Terminal Coding Agent** [GitHub] [Live Website] [2026 - Present]
 - Building an open-source **terminal-based coding agent** designed to let developers write code, manage development sessions, and interact with an AI-assisted coding workflow directly from the terminal.
 - Architected the project as a modular **monorepo** separating the **CLI, server, database, and shared** packages, enabling independent development of the developer interface, backend services, persistence layer, and shared abstractions.
 - Designed the product around a **terminal-first developer experience**, eliminating browser/Electron dependencies while supporting persistent coding sessions and customizable terminal themes.
 - Implemented a **client-side distributed tool execution runtime** where the backend streams tool-call specifications (file reads, grep/glob searches, shell commands) but all execution runs locally within strict `process.cwd()` path-confinement boundaries, ensuring proprietary source code never leaves the developer's machine.
- **ControlPlane.ai - Real-Time AI Governance Proxy** [GitHub] [2026]
Accenture Innovation Challenge 2026 (Team Leader)
 - Led the team for the **Accenture Innovation Challenge 2026** (Problem Statement 1: "Reinvent with AI"), architecting a sub-10ms reverse-proxy AI governance layer between client applications and LLMs.
 - Engineered a synchronous **fast path in Rust (<10ms)** using Hyper and lock-free policy caching (`arc-swap`) for regex/entropy secret detection, tiered cost cap enforcement (block, escalate, edit), retry-loop detection, and session risk tracking.
 - Implemented an asynchronous **shadow path (<2s)** with 13 automated checks spanning DeepEval hallucination scoring, NLI groundedness, Presidio PII detection, LLM Guard toxicity/bias filtering, and 3-layer prompt injection detection.
 - Designed a resilient architecture with a **fail-open guarantee**, a SHA-256 hash-chained audit ledger in PostgreSQL, an escalation queue for human review, and a real-time Next.js 16 SSE dashboard.
- **BlindDrop - Privacy-Focused File Transfer Portal** [Project Link] [2026]
 - Built a privacy-focused **anonymous file-transfer system** enabling users to upload files without authentication, receive one-time 6-character download codes, and automatically delete files after download or expiry.
 - Architected a **Next.js + TypeScript + Tailwind CSS frontend** with a **Flask REST API backend**, implementing upload/download workflows, admin dashboards, authentication, audit logs, transfer statistics, and database-management interfaces.
 - Implemented **hash-based horizontal sharding across three SQLite databases** using `MD5(sessionID) % 3`, with shard-aware query routing, scatter-gather token lookup, integrity verification, and dedicated sharded/unsharded table management.
 - Built secure file lifecycle management using **UUID-based storage, SHA-256 integrity checks, configurable expiry, background cleanup, one-time token consumption, rate limiting, security headers, and no-store caching**, supporting files up to 100 MB.
- **Algorithmic Trading Bot - Upstox** [Project Link] [2024 - 2025]
 - Developed a modular **Python-based algorithmic trading bot** for live trading on the **Upstox API**, separating account connectivity, market data, strategy generation, order execution, and trade orchestration.
 - Implemented dedicated modules for **historical/live market data, trading signals, order execution, option expiry and instrument discovery, and automated trade management**.

- Built configurable risk controls for **stop-loss, take-profit, maximum trades per day, and maximum daily loss**, with strategy modularization supporting algorithms such as EMA crossover.
- **Adani FinTell Suite - Enterprise Invoice & Compliance Platform** [\[GitHub\]](#) [Finnovate Hack 2025]
IIT Gandhinagar & Adani Enterprises (Team Leader & Runner-Up)
 - Led the runner-up team at the **Adani Finnovate Hackathon 2025**, building an AI-powered financial intelligence and automated invoice auditing platform for enterprise procurement.
 - Built an intelligent document processing pipeline using **Google Gemini AI**, achieving **80%+ extraction accuracy** across multi-format invoices (PDF, PNG, JPEG) with single and bulk upload capabilities (up to 50 invoices).
 - Implemented end-to-end compliance verification including **real-time GST portal verification**, HSN/SAC regulatory tax rate validation, arithmetic accuracy checks, and AI-derived price anomaly benchmarking against market rates.
 - Developed interactive analytics interfaces including a processing pipeline timeline modal, an anomalies dashboard for duplicate and fraud detection, and a conversational AI analytics assistant.
- **Machine Learning Projects**
 - **Poetic Text Generator** [\[Project Link\]](#)
 - * Developed an RNN-based **LSTM text-generation model** for generating poetic text from sequential training data.
 - * Implemented **character-level tokenization and sequence modeling** to learn relationships between characters and generate new text.
 - * Added **temperature-controlled sampling** to control generation randomness and produce varied outputs.
 - **Plant Disease Detection Model** [\[Project Link\]](#)
 - * Developed a **CNN-based multi-class image classification system** for detecting plant diseases from leaf images.
 - * Trained the model using the **PlantVillage dataset** with **TensorFlow and Keras**.
 - * Built the classification and evaluation pipeline to identify disease categories and assess model performance across classes.
 - **Generating Handwritten Images** [\[Project Link\]](#)
 - * Implemented a **Deep Convolutional Generative Adversarial Network (DCGAN)** for generating handwritten digit images.
 - * Developed separate **generator and discriminator architectures** using TensorFlow and Keras.
 - * Trained the model on the **MNIST dataset** and evaluated progressive image synthesis to generate realistic digit samples.
 - **Customer Churn Prediction Model** [\[Project Link\]](#)
 - * Developed a **customer churn prediction system** using supervised machine learning on structured customer data.
 - * Applied **SMOTE** for class balancing and trained a **Random Forest** classifier to predict customer churn.
 - * Serialized preprocessing encoders and model components to support **real-time inference** on new customer records.
 - **Autism Prediction Model** [\[Project Link\]](#)
 - * Developed a machine-learning-based **ASD prediction system** using structured healthcare-related features.
 - * Evaluated **Decision Tree, Random Forest, and XGBoost** models with SMOTE-based class balancing.
 - * Applied **data preprocessing, exploratory data analysis, hyperparameter tuning, cross-validation, and model evaluation** to compare model performance.
 - **Twitter Sentiment Analysis Model** [\[Project Link\]](#)
 - * Developed a **Logistic Regression-based NLP classifier** to classify tweets as positive or negative.
 - * Built a text-classification pipeline involving **text preprocessing and feature representation** for supervised learning.
 - * Achieved **77.67% test accuracy** on the sentiment classification task.

WEB DEVELOPMENT PROJECTS

- **Boeing BUILD 2026 Website** [\[Live Website\]](#) [2026]
 - Designed and developed the official **Boeing BUILD 2026 India** website, translating the program's requirements into a polished public-facing digital experience.
 - Built responsive interfaces for **program information, event details, application workflows, schedules, and participant-facing content**, ensuring consistent usability across desktop and mobile devices.
 - Managed the implementation and delivery of the website end-to-end, collaborating with stakeholders to translate requirements into production-ready web functionality.
- **Research Park Website - IIT Gandhinagar** [\[Live Website\]](#) [June '25 - December '25]
 - Designed and contributed to the official **IIT Gandhinagar Research Park** website, presenting its industry-academia collaboration ecosystem, infrastructure, services, partners, and success stories.
 - Developed responsive interfaces for sections covering **industry partnerships, consultancy projects, laboratory infrastructure, faculty collaboration, internships, IP transfer, and events**.
 - Focused on **navigation, institutional branding, responsive design, and performance**, delivering a polished web presence for an institutional research and innovation platform.
- **Chess Website of IIT Gandhinagar** [\[Project Link\]](#) [December '24 - February '25]
 - Developed and maintain the official web platform for the **IIT Gandhinagar chess community**, consolidating campus chess events, announcements, registrations, and resources.

- Implemented dedicated workflows for **event listings, competition registration, announcements, gallery management, and contact information**.
- Built the frontend using **HTML, CSS, and JavaScript**, including interactive form handling and dynamic event updates with a responsive, user-friendly interface.
- **Zero Waste Gujarat Website** [Live Website] [2024 - Present]
 - Designed and developed the website for **Zero Waste Gujarat**, a textile waste-to-value initiative focused on recycling discarded fabrics into recycled fiber, sustainable nonwovens, eco-friendly packaging, and carbon-credit solutions.
 - Built the platform using **Next.js** with **server-side rendering (SSR)** to support SEO, performance, and discoverability.
 - Implemented responsive interfaces communicating the organization's **textile collection, processing, manufacturing, carbon-credit, and sustainability initiatives**.
- **ToonifyIt - JSON to TOON Converter** [Live Website] [2025 - Present]
 - Built and launched an independent **LLM developer tool** that converts JSON to **TOON (Token-Oriented Object Notation)** for more token-efficient representation of structured data in LLM workflows.
 - Developed a real-time, browser-based conversion engine supporting **JSON ↔ TOON conversion**, configurable delimiters and indentation, length markers, token-count comparison, and copy/download workflows.
 - Designed the application around **client-side processing and privacy**, keeping user data on-device while supporting structured data such as tabular datasets, API responses, and large JSON payloads for LLM prompts.
- **Vardhaman Agencies Website** [Live Website] [2024 - 2025]
 - Built the official B2B/B2C website for **Vardhaman Agencies**, a packaging-solutions business serving customers across India.
 - Developed product-focused interfaces for **bubble wrap rolls, stretch films, custom packaging solutions, product discovery, and quotation workflows**.
 - Implemented a responsive frontend with **React.js**, modular components, product navigation, search functionality, and mobile-friendly layouts.

SKILLS SUMMARY

- **Languages:** Python, C, C++, Java, Rust, SQL, JavaScript, TypeScript, HTML, CSS
- **Frameworks & Libraries:** React.js, Next.js, Tailwind CSS, Angular, RxJS, Spring Boot, FastAPI, Axum, Flask, Google ADK, NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch
- **AI / ML / LLM:** AI Governance, Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, Agentic AI, Prompt Engineering, Gemini, Vector Retrieval, AI Agents
- **Databases & ORMs:** PostgreSQL, MongoDB, MySQL, SQLite, SQLx, Prisma
- **Cloud & Infrastructure:** Google Cloud Platform, Cloud Run, Cloud Build, Cloud SQL, Vertex AI, Docker, GCP IAM
- **Security & Systems:** Authentication, Authorization, RBAC, PII Masking, Prompt Injection Detection, SQL AST Validation, BYOK, REST APIs, Microservices, SSE, API Security
- **Tools:** Git, GitHub, Linux/Unix Shell, Bash, VS Code, Jupyter Notebook, Google Colab, MATLAB, Postman
- **Quantitative / Trading:** Algorithmic Trading, Backtesting, Live Trading, Brokerage APIs, Trading Strategies, Risk Management
- **Interpersonal Skills:** Teamwork, Technical Leadership, Cross-Functional Communication, Problem Solving

RELEVANT COURSES

- **Computer Science Core:** Data Structures and Algorithms I (ES 242), Operating Systems (CS 330), Computer Networks (CS 331), Databases (CS 432), Compilers (CS 327), Theory of Computing (CS 201), Digital Systems (ES 204), Software Tools and Techniques for CSE (CS 202), Computing (ES 112)
- **AI, Data Science & Cognition:** Foundations of AI: Multiagent Systems (CS 329), Introduction to Data Science (CS 328), Software Tools & Techniques for AI (CS 203), Data-Centric Computing (ES 113), Computation and Cognition (CG 501)
- **Mathematics & Statistics:** Discrete Mathematics (ES 214), Probability, Statistics, and Data Visualization (ES 114), Calculus of Single Variable and Linear Algebra (MA 103), Calculus of Several Variables (MA 205), Ordinary Differential Equations (MA 104), Numerical Methods (MA 203)
- **Management, Economics & Engineering:** Business, Organization & Finance (MS 491-XIII), Laboratory in Entrepreneurial Motivation (MS 491-VIII), International Marketing (MS 492-VII), Neuromarketing (MS 404), Economics (HS 151), Competitiveness and Ethical Leadership (HS 392), Design, Innovation, and Prototyping (ES 115), Bionanotechnology (BE 303)

ACHIEVEMENTS

- Team Leader & Runner-Up, **Adani Finnovate Hackathon 2025** (IITGN & Adani Enterprises): Led a 4-member team to build an AI-powered enterprise invoice compliance platform with real-time GST validation and Gemini OCR.
- Team Leader, **Accenture Innovation Challenge 2026** (Problem Statement 1: "Reinvent with AI"): Architected ControlPlane.ai, a sub-10ms dual-path AI governance proxy.
- Awarded **Dean's List** for academic excellence in Semester I (Academic Year 2023-24) at IIT Gandhinagar.
- Secured **All India Rank 1505** (top 1% out of 1,80,372+ candidates) in **JEE Advanced 2023** to pursue Computer Science and Engineering at IIT Gandhinagar.
- Secured **99.65 percentile** in **JEE Main 2023** among more than 1.2 million candidates.

- **Revive & Thrive:** Achieved Rank 2 in the institute-level competition at IIT Gandhinagar.
- Qualified **NTSE Stage 1** (National Talent Search Examination) from Haryana State.
- Awarded multiple gold and silver medals in district- and state-level **SOF Olympiads**.
- Top Ranks in National Aptitude & Scholarship Tests: **AIR 3** in FTRE, **AIR 6** in FIITJEE Big Bang Edge, **AIR 11** in VMC Entrance Exam, and **AIR 54** in Tallentex.
- Garnered **10,000+ followers and 9.2M+ views on Quora**, authoring analytical content on mathematics, chess algorithms, and problem-solving.
- Generated over **100K organic impressions** via SEO and data-driven editorial strategies on a self-published finance blog with a primary US/UK readership.

POSITIONS OF RESPONSIBILITY

- **Senior Tech Team, Entrepreneurship Initiative, IIT Gandhinagar** [2026 - Present]
 - Contributing to technical initiatives, digital infrastructure, and technology-driven activities within IIT Gandhinagar's entrepreneurship ecosystem.
- **Web Development Team, Tinkerers' Lab, IITGN** [2025 - Present]
 - Contributed to development and maintenance of web infrastructure and digital platforms supporting IITGN's student maker and technology community.
- **Webmaster, Student Academic Council, IITGN** [July '25 - Present]
 - Maintained and regularly updated the official Student Academic Council website, ensuring good website speed, reliable functionality, and a smooth user experience.
- **Senior Marketing Executive, EII, IITGN** [October '24 - February '25]
 - Increased outreach by optimizing social media strategies and engaging a broader audience.
 - Led the LinkedIn team to enhance the EII's digital presence, successfully increasing engagement and reach.
- **Senior Events Executive, Amalthea, IITGN** [April '24 - November '24]
 - Organized pre-events to promote chess within the IIT Gandhinagar campus, driving interest and participation in the larger Amalthea summit.
 - Managed logistics and collaborated with multiple teams to ensure the smooth execution of events.
- **Organizer, Kautilya Chess League, IITGN** [March '24 - April '24]
 - Successfully organized a week-long chess league involving more than 80 players across 6 competitive teams.
 - Fostered a culture of intellectual engagement and sportsmanship, attracting participants from diverse backgrounds.
- **Organizer, Triunity Chess, Hallabol, IITGN** [February '24]
 - Introduced a unique 3-player chess variant to foster innovative thinking and promote chess culture within the campus.
 - Successfully engaged students through this novel approach, enhancing interest in strategic gameplay.
- **Founder, Finimoney.com** [June '23 - May '24]
 - Founded and managed an independent finance blog targeting a US/UK readership, applying data-driven editorial and SEO strategies to grow an international audience.
 - Developed domain expertise in personal finance, investment analysis, and content monetization through consistent long-form publishing.
- **Marketing Team Member, Amalthea, IITGN** [September '23 - April '24]
 - Collaborated with a team to execute outreach strategies targeting students and industry professionals to increase participation in the event.

EXTRA-CURRICULAR ACTIVITIES

- Playing Chess
 - FIDE Rating: 1437 (Rapid)
 - Chess.com Rating: 1938 Rapid (Peak)
- Competitive Programming
 - Codeforces Rating: 1200+ | [Codeforces ID](#)
 - LeetCode Rating: 1500+ | [LeetCode ID](#)

COURSES AND CERTIFICATIONS

- SEBI - Investor Certification Examination (issued by National Institute of Securities Markets (NISM))
- SEO Training Masterclass 2024: Beginner To Advanced SEO (issued by Udemy)
- Finlatics Certified Investment Banking Experience Program (issued by Finlatics)
- Business and Finance (issued by IBM)
- Mathematical Optimization for Business Problems (issued by IBM)
- What Is Generative AI? (issued by LinkedIn)
- What Is Business Analysis? (issued by LinkedIn)
- Ashurst UK - Building Your Personal Brand Job Simulation (issued by Forage)
- BCG - Introduction to Strategy Consulting Job Simulation (issued by Forage)
- White & Case - Project Finance Job Simulation (issued by Forage)
- HSBC - Commercial Banking Job Simulation (issued by Forage)
- Leadership Skills (issued by Coursera)
- Pre-MBA Statistics (issued by Coursera)